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Project Proposal 03 — Political Blogs (Adamic & Glance 2005)

Analisi e Visualizzazione delle Reti Complesse — Network Science module, final project.

Abstract

Adamic and Glance (2005) crawled the U.S. political blogosphere two months before the Bush–Kerry election. The resulting network has **1 490 blogs joined by 19 090 directed hyperlinks**. Each blog is hand-labelled as *liberal* or *conservative*.

This is the textbook empirical example of a polarised online ecosystem. The paper's headline finding — that **91% of links are within-side and only 9% cross-side** — is the single most-cited homophily statistic on a public network dataset.

The project does three things:

1. **Replicates** the Adamic–Glance polarisation result.
2. **Identifies** the structurally influential blogs and the rare cross-cutting brokers between the two camps.
3. **Tests** — through a directed configuration-model null — whether the observed segregation exceeds what chance alone produces.

1. Introduction and motivation

Adamic and Glance's paper is one of the most cited in computational social science. The reason is simple: the dataset combines three properties that rarely show up together.

1. **Structural clarity.** Two almost-disconnected blobs joined by a thin bridge, visible to the naked eye on any decent layout.
2. **Ground truth.** Every blog was hand-coded as liberal or conservative, so algorithmic partitions can be compared against a real-world categorisation.
3. **Directedness.** Blog A links *to* blog B, so in-degree and out-degree mean different things. The link-analysis toolbox of **Lecture 19** (PageRank, HITS) becomes the natural way to measure influence.

Choosing this project means picking up a story that any reader understands immediately: U.S. political polarisation at the dawn of the political-blogs era. In exchange, you take on the methodological discipline directed graphs demand. Every analysis must state whether it was run on the directed graph, the undirected projection, or the largest strongly connected component. **Sloppy projections are the most common source of bugs on this dataset.**

The reward is a report that reads like a small piece of political science with network-science methodology underneath.

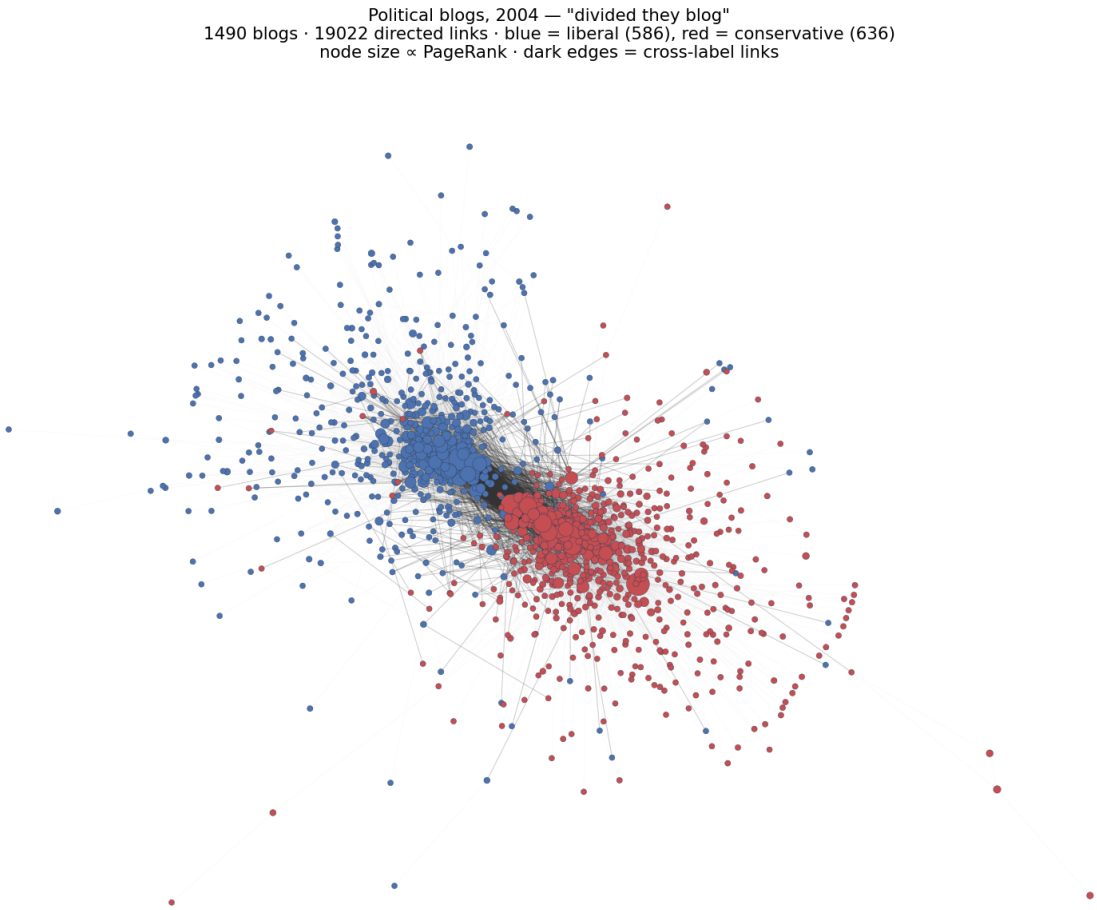

Figure 1: Hero figure: the political blogosphere in 2004 — “divided they blog”. Blue = liberal, red = conservative; node size is proportional to PageRank; dark edges highlight cross-label links.

2. The dataset

2.1 Source and provenance

Collected by Lada Adamic and Natalie Glance for *The political blogosphere and the 2004 U.S. election: divided they blog* (LinkKDD, 2005). Distributed as `polblogs.gml`. Mark Newman historically hosted a copy at http://www-personal.umich.edu/~mejn/netdata/polblogs.zip; KONECT mirrors a stripped version at http://konect.cc/networks/moreno_blogs/.

2.2 Structure and semantics

- **1 490 nodes**, **19 090 directed edges**.
- Each node is a single blog and carries two attributes:
  - `label`: the blog URL (e.g. `dailykos.com`), which makes the graph human-readable.
  - `value`: the hand-coded political orientation — **0 = liberal**, **1 = conservative**.
- A directed edge `A` → `B` means blog `A` contains at least one hyperlink to blog `B`.

2.3 Caveats

- The graph is **not** strongly connected.
- Self-loops and duplicate edges exist in the raw file and must be removed before analysis.
- For path-based metrics use the **largest strongly connected component**. For most other analyses the **largest weakly connected component** is appropriate — but state the choice explicitly for every number.
- Ground-truth labels are hand-coded and not infallible. Disagreements between algorithm and label are interpretive opportunities, not errors to hide.

3. Research questions

The project should commit to **three or four** of the following. The questions form a narrative arc: *how polarised is the graph, who rules each side, who bridges, and is the polarisation more than chance can produce?*

1. **How strong is political homophily?** Adamic and Glance's headline — **91% within-side, 9% cross-side** — is the cleanest documented homophily effect on a public dataset, and the first result any group on this project should reproduce. But the aggregate ratio hides variation at the node level: some blogs sit comfortably inside their camp and link only to allies; others are consistent cross-cutters. The **E-I (external-internal) index** quantifies this per node. Plotted as a distribution per side, it reveals whether the two camps are equally insular or asymmetric. Homophily is the core material of **Lecture 5**, and this dataset is the teaching example for it. *Tools:* within- vs cross-label edge counts; attribute assortativity on `value` (**Lecture 5**); the 2×2 mixing matrix; per-node E-I index with its distribution per side.

2. **Who are the most influential blogs, and does influence differ by side?** In a directed graph, in-degree and out-degree say different things. A blog with high **in-degree** is one that many others cite; a blog with high **out-degree** is one that cites many others. PageRank and HITS give recursive versions of these ideas — the course develops them in **Lecture 19**. The top-20 blogs under PageRank on this dataset include names that any reader of U.S. political media in 2004 will recognise instantly (`dailykos.com`, `instapundit.com`, `andrewsullivan.com`, `atrios.blogspot.com`) — a reassuring face-validity check that the methodology is returning sensible answers. The more interesting structural question is whether the two sides are *symmetric* in their authority structure. Does each side host a handful of dominant blogs and a long tail, or does one side have a flatter internal hierarchy than the other? Asymmetry is itself a documented Adamic–Glance finding and is worth replicating. *Tools:* in- and out-degree centrality (**Lecture 6**); PageRank and HITS hubs/authorities (**Lecture 19**); top-20 tables side-coloured by label.

3. **Are the in- and out-degree distributions heavy-tailed?** A dataset with an obvious "superstar" structure naturally raises the power-law question. In-degree on the political blogs shows a heavier tail than out-degree: a small number of blogs receive thousands of links, while most blogs emit only a handful. Whether either tail is technically a power law (**Lecture 7**) or merely power-law-like is a secondary question. The substantive point is that the two distributions differ in shape, and the shape difference has an intuitive interpretation: **most bloggers follow a few opinion leaders; few bloggers are opinion leaders.** *Tools:* log-log CCDFs of in-

and out-degree on shared axes; tail fits via the `powerlaw` package with a log-normal comparison (**Lecture 7**).

4. **Do community-detection algorithms recover the left–right split?** This is the question for which the Political Blogs dataset was invented. Because every node carries a ground-truth label, a community-detection algorithm's partition can be compared against an external categorisation — the situation that **Lecture 15** treats as the gold standard of evaluation. The test is demanding: the algorithm is given only the graph, and must recover a bipartition that aligns with a political category it has no direct access to. Typical results:

   - **Louvain or Leiden** on the undirected projection achieves NMI above 0.65 against the labels.
   - **Infomap** (**Lecture 14**) on the directed graph is often competitive or slightly better, because flow-based community detection respects link direction.

   The cases where algorithm and label disagree — a handful of blogs per run — are the single most interesting thing to discuss in the report. *Tools:* Louvain or Leiden on an undirected projection; Infomap on the directed graph; comparison with ground-truth labels via NMI, AMI, ARI (**Lecture 15**); explicit enumeration of disagreement cases.

5. **Are the two sides structurally symmetric?** The within-side structure is an empirical question the dataset can answer. Restrict to the induced subgraph of all liberal nodes, then of all conservative nodes. Four numbers describe each subgraph — density, clustering, reciprocity, average in-degree — and they are either similar (symmetric blogosphere) or systematically different (asymmetric). On this dataset the conservative side is typically somewhat **denser, more reciprocal, and more tightly clustered**. That is the kind of asymmetry that invites cautious substantive interpretation — and caution is the key. *Tools:* induced subgraphs per side; density, average clustering, reciprocity, average in-degree on each (**Lectures 2–4**).

6. **Who are the bridges between the two camps?** The structural-holes framework of **Lecture 4** is almost tailor-made for this question. In a polarised graph, most blogs sit entirely inside their camp. A small number sit at the edge and post cross-cutting links. These rare cross-cutters — "who links outside their own side and by how much" — are the brokers, and they are **not** the hubs. Hubs are typically the most insular nodes, precisely because their audience is inside one camp. Identifying the five to fifteen cross-cutting brokers, naming them, and discussing what their links look like turns the structural analysis into a narrative. *Tools:* fraction of cross-label out-links per node; ranked list; manual inspection of the top cross-cutters.

7. **Does the observed segregation exceed chance?** An observed 91/9 split looks dramatic, but a sceptical reader is entitled to ask whether the split is more than one would expect from the *degree distributions alone*. A directed configuration model preserves each node's in- and out-degree while randomising edge endpoints. Counting cross-label edges in 50 such randomisations produces a null distribution against which the observed count can be placed. The expected result is that the real graph's cross-label count falls far into the tail of the null distribution with a p-value effectively zero. In plain language: **the observed polarisation is stronger than degree heterogeneity alone can explain** — the two camps choose their links, not just their volumes. *Tools:* directed configuration model preserving in- and out-degree (**Lecture 8**); cross-label edge counts over ≥ 50 realisations; p-value of the observation against the null.

8. *(optional)* **How fast does a cascade leak across the political divide?** A linear-threshold or independent-cascade simulation (**Lecture 18**) seeded inside one camp turns structural polarisation into information dynamics. How long does it take for the cascade to leak to the other side? How large is the leak? This is an elegant bridge from static structure to dynamical behaviour and a natural complement to the structural questions above.

4. Expected findings

A well-executed project should reach, at minimum, the following results:

- Replication of the Adamic–Glance headline: approximately **91% of edges within-side, 9% cross-label** in the 2×2 mixing matrix. The attribute assortativity coefficient on `value` is positive and strong, consistent with one of the most severe homophily signals documented on a public dataset.
- An in-degree tail that is heavier than the out-degree tail: a handful of very-linked-to blogs (`dailykos.com`, `instapundit.com`, `andrewsullivan.com`, `atrios.blogspot.com`) and many blogs with few outgoing links. Fitted exponents near **2.1 for in-degree** and **2.7–3.0 for out-degree** are typical.

- The conservative induced subgraph is typically **denser and more reciprocal** than the liberal one — an Adamic–Glance finding worth replicating.
- A community partition of **2–3 dominant communities**; NMI against ground-truth labels above **0.65**; Infomap results comparable or slightly better due to flow-sensitivity.
- Directed configuration-model null in which the expected cross-label edge count is roughly **an order of magnitude larger** than the observed count; observed p-value effectively zero.
- A short list of **5–15 cross-cutting broker blogs** (those with > 20% cross-label out-links).

5. Visualization plan

Visual design is the group's responsibility — the dataviz module covers the *how*. This section only states the *what*: for each finding the project claims, a visualisation should carry the corresponding message.

- **The polarised structure.** The report's *hero figure*: a single, carefully designed view that makes the two-camp segregation immediately visible while exposing the small number of cross-cutting edges.
- **"How strong is the homophily?"** A visualisation that communicates the imbalance between within-label and cross-label edges in a single reading.
- **"Who are the influential blogs, and are the two sides symmetric?"** A visualisation that shows the top-ranked blogs per centrality, colour-coded by side, so the reader can see whether influence is distributed symmetrically.
- **"How do in- and out-degree distributions compare?"** A distributional view that puts both tails side by side.
- **"Do algorithmic communities match the ground-truth labels?"** A visualisation of the partition-vs-label agreement at the community level.
- **"Is the observed segregation beyond chance?"** A visualisation placing the observed cross-label edge count against the null-model distribution.

An interactive version is particularly compelling here because blogs are nameable. Pick one interaction task — "hover to reveal a blog's URL and top neighbours", "filter the graph by label or community", "search for a specific blog" — and do it well.

6. Possible extensions

- **Text-plus-structure.** Pair the graph with scraped post content from the Internet Archive and ask whether structurally-bridging blogs also bridge topically.
- **Modern comparison.** A small hand-collected 2020s political network (Twitter, Bluesky, Mastodon) analysed with the same pipeline supports a "has polarisation changed?" appendix.
- **Signed perspective.** Hand-label 100 cross-label edges as approving versus critical and rerun the analysis under signed assumptions — a direct use of the structural-balance framework of **Lecture 5**.
- **Stochastic block model.** Fit a 2-block SBM using the ground-truth labels as blocks and compare the fitted edge probabilities with the empirical mixing matrix (**Lecture 15**).
- **Geographical view.** Map blogs to U.S. states (recoverable from public archives for most state-branded blogs) and explore the spatial distribution of ideological lean — a direct use of the cartographic material from the dataviz module.

7. Common pitfalls

- **Directedness discipline** is the number-one source of bugs. For every metric, state whether you used the directed graph, the undirected projection, or the reciprocity-only projection.
- **Path-based metrics on the whole graph** (rather than on the largest strongly connected component) produce undefined behaviour.
- **The raw GML contains duplicate edges** that crash vanilla parsers — plan for a robust loader.
- **Ground truth is not gospel.** Report the NMI *and* enumerate disagreements.

References

- Adamic, L. A. and Glance, N. (2005). *The political blogosphere and the 2004 U.S. election: divided they blog.* LinkKDD.

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- Page, L., Brin, S., Motwani, R., and Winograd, T. (1999). *The PageRank citation ranking.* Stanford InfoLab.
- Rosvall, M. and Bergstrom, C. T. (2008). *Maps of random walks on complex networks reveal community structure.* PNAS.
- Newman, M. E. J. (2003). *Mixing patterns in networks.* Physical Review E.

- Course syllabus and lecture slides: `netsci/ns_syllabus.md`, `netsci/pdfs/`.